

Unequal Job Opportunities and the Measurement of Welfare Inequality

Hisham Haydar

LISER and University of Luxembourg

France 2016 · EU-SILC priced through EUROMOD

Two people, the same job, different reasons

Person A

40 hours

same occupation

bottom wage quintile

chose it from a wide set

Person B

40 hours

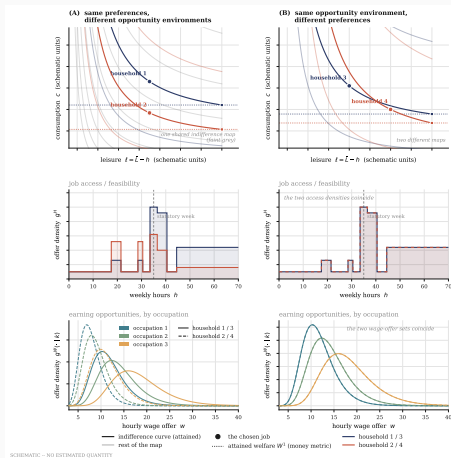
same occupation

bottom wage quintile

had almost nothing else

Whatever the choice set cannot express,
the taste parameter absorbs.

What the model has to separate



Schematic. Stylised parameters; no estimated quantity on any axis.

Why random opportunities, not a common choice set

	common choice set	random opportunities
what varies across households	the budget only	the whole offer density
unexplained variation goes to	preferences	job access, earning opportunities, preferences
welfare comparison available	levels at a common set	levels at own set
what is identified	tastes	tastes <i>and</i> availability

Remove the occupation term and the effect does not vanish —
it relocates.

tertiary-education loading moves by six standard errors

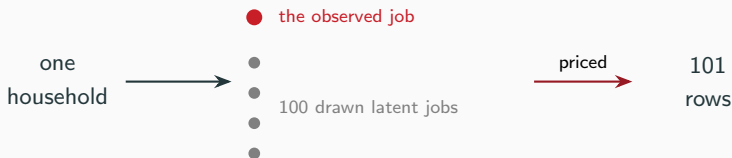
What the opportunity distribution is made of



estimated jointly with Box–Cox **preferences** in a single likelihood

The hours factor carries no household-specific coordinate.

Data and the sampled-job estimator



EU-SILC / SRCV 2016 · 1,555 households · 101 alternatives each · 157,055
priced rows

conditional logit with the sampling-of-alternatives correction

Every row is priced through the tax–benefit system.

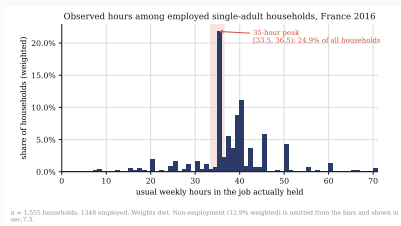
Consumption is computed, not surveyed; the mapping is deterministic.

Identification: what is pinned, what stays conditional

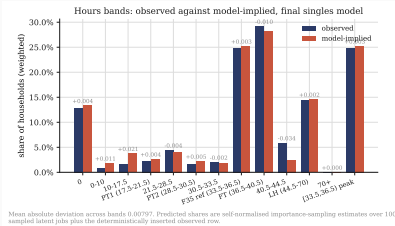
what the data pin	what stays conditional
the four availability factors	access <i>versus</i> capability
the preference block	persistent household heterogeneity
the statutory-week peak	

The decomposition is structural and model-conditional. It is not causal.

Hours, and the statutory week



observed hours

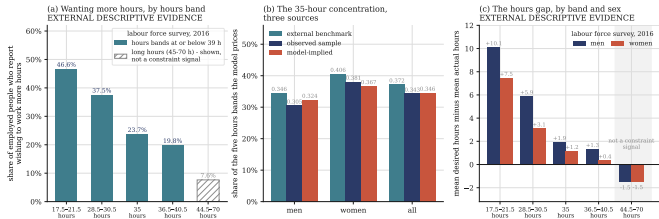


observed against model-implied

statutory band 24.85% observed · 25.20% model-implied · mean absolute error 0.0083 over 12 bins

employment 0.8708 / 0.8668 · occupation largest deviation 0.0070 · bottom offered-wage quintile over-predicted by 5.3 points

External validation



All three panels are external evidence about hours. No moment from this source enters the model's objective, and some of it identifies a parameter. The population is all persons aged 20-60 with a reported usual hours figure (43,767 records); the estimation sample is single-adult households, so composition differs by construction and no gap between them is a prediction error. Panel (a). Sexes pooled, because the wish-to-work more question was returned pooled. Panels (a) and (c) are EXTERNAL DESCRIPTIVE EVIDENCE ONLY: the structural model has no desired hours or underemployment outcome, so these moments are not compared with a model prediction, and are not used to validate an estimated offer-weight parameter, and identify nothing in the model. No model quantity appears on either axis of either panel, and no rank statistic against an estimated offer weight is computed or shown. Panel (a) is coded from the survey's wish-to-work-more variable as 1 = No, 2 = Yes per the published variable codebook (version of 8 July 2021); the delivered export's own field name had the map the other way round and is superseded. Across the five bands 20.3% of employed people wish to work more hours. The long hours band is drawn but marked, because a band can be thin as OFFERS while the people already working 45-70 hours are the least likely to want more. Panel (b) is the validation panel and does not use the wish-to-work-more variable. All three sources are put on the share of the five bands the model prices, the only base on which they are commensurable. Panel (c) is mean desired hours minus mean ACTUAL hours within the same band, both weighted. The gap is positive on every band at or below 39 hours for both sexes and turns negative only at long hours. It is larger for men than for women in all four short-and-medium bands.

statutory band 37.2% external · 34.3% observed · 34.6% model-implied

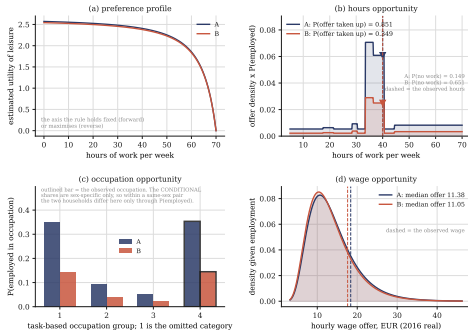
The welfare measure, and two real households

Same estimated preferences, same observed job, most different estimated opportunities

A: male, 35-39, high education, no children, intermediate

B: male, 40-44, low education, no children, rural

matched on the observed job: both employed, occupation 4, hours band "H", wage quintile 4. d_pref = 0.021 (admissible median 0.783). d_opp = 0.502 (admissible median 0.209).

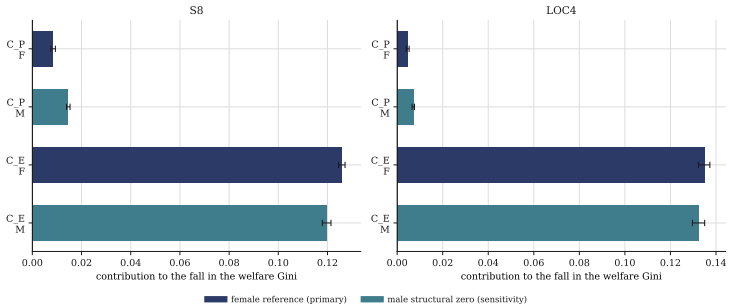


Final singles model, corrected frame: the pair is the argmax of the stated rule over 19,116 admissible employed pairs, not a hand pick. Panels (b)-(d) are the model's OFFER distribution, not the choice distribution. In the final singles model the hours band index carries no household-specific coordinate, so the two hours curves differ only by the scalar P(employed); the model lets opportunity differ across households in the participation margin, the sex-specific occupation shares and the wage offer location, and nowhere else. Terminology: B.227 sec.12 - job access / feasibility, earning capacity / wage-offer technology, preferences. No household identifier is used.

same preference profile (4.61 and 4.56) · employment probability **0.851** against **0.349**

The headline

Preference and environment contributions, both reference conventions (raw welfare)



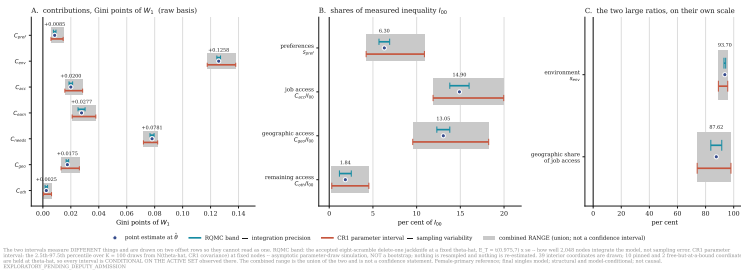
Owen value for the partition $\{ \{ \text{preferences} \}, \{ \text{job access, earning opportunities, endowments-needs} \} \}$ on $v(S) = I(\text{baseline}) - I(S)$. Bars are SIGNED and are never renormalised. Error bars are the eight-scramble delete-one jackknife at $t = 2.3646$. F = female reference (primary), M = male structural zero (sensitivity); the two are NEVER averaged.

FINAL_SINGLES_PROVISIONAL_PENDING_ECONOMICS_REVIEW

The environment $93.70 \pm 0.61\%$ · Preferences
 $6.30 \pm 0.61\%$

The headline

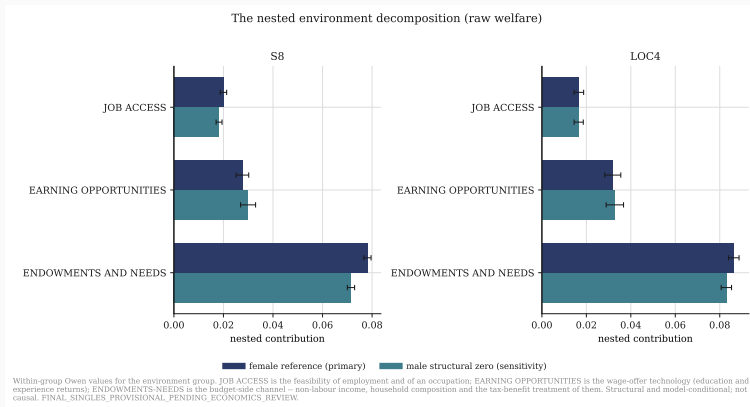
The headline decomposition under two kinds of uncertainty, kept apart



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 $6.30 \pm 0.61\%$

parameter interval: preferences [4.22, 10.89] the environment [89.11, 95.78]

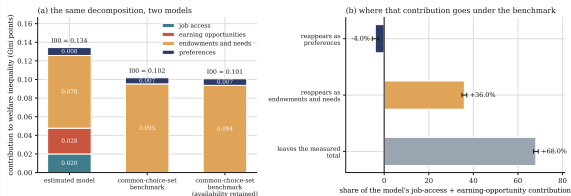
Inside the environment



Household endowments and needs $58.18 \pm 1.47\%$ · Earning opportunities
20.62 $\pm 1.63\%$ · Job access 14.90 $\pm 1.10\%$
market side together: $35.51 \pm 0.95\%$

What a conventional model would have found

Welfare-inequality decomposition: estimated model against the common-choice-set benchmark



Single-adult households, France 2010. Money-metric: welfare level W1, female reference, raw basis. Own values on the grouped game: bands are the eight-scramble jackknife at $t = 2.364624251$. FINAL_RUM_BENCHMARK_PROVISIONAL_PENDING_ECONOMICS_REVIEW

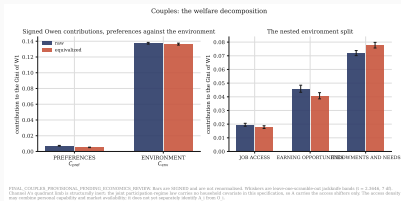
preferences share 6.4% against 6.3% · measured inequality -24.2% raw · sex
leisure gap +0.428 → -1.991

What is fragile, and what is not

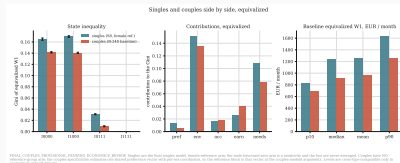
perturbation	preferences share	the environment side
reference-preference convention	6.30 $\rightarrow$ 10.89%	< 3.5%
a bound on a preference curvature	$\approx$ 13% move	< 3.5%
pinning the couples male leisure block	up to 73% move	< 3.5%

No sign and no ordering changes anywhere. Market-side total 35.51% against 35.88% across references.

Couples



the couples decomposition



singles and couples, equivalized

2,275 couples · same nested order, same the environment dominance

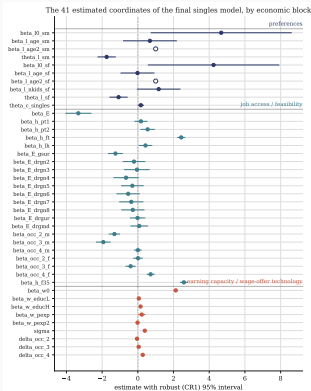
The couples preferences share is not structurally robust. Singles remain the quantitative headline.

- **The method.** A household-specific opportunity distribution estimated jointly with preferences, every job priced through the real tax–benefit system.

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- **The method.** A household-specific opportunity distribution estimated jointly with **preferences**, every job priced through the real tax–benefit system.
- **The finding.** The environment carries 93.70% of measured money-metric welfare inequality; inside it the budget side carries more than the market side.
- **The margin.** The **preferences** share is the fragile quantity: 6.30% to 10.89% across the two normative conventions.

B1 — The estimated coefficients

[illegible]

41 estimated structural parameters, robust 95% intervals

B2 — Exhaustiveness: how the decomposition was made to close

defect	what fixed it
a missing budget channel	added as a fourth player from its own marginals
asymmetric money-metric inversion	reference side moves with the coalition
an incomplete male reference block	structural zero on the child leisure term

Exhaustiveness here is a tested property, pre-stated before any cell was read.

the fully common state is 0.000000 — numerically zero

B3 — Equivalization, and why the scale has to move

	raw	equivalized
preferences	6.30%	7.98%
the environment	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing each household's own scale leaves a residual of 0.071799 — precisely the weighted Gini of one over the scale.

B4 — Why the hours-wish evidence is context, not a moment

ground	what it shows
definitional	the candidate's optimum tracks pay: 98.1% pay more, median 2.58×
identification	the hours block moves the moment by exactly 0
numerical	24.3% against 20.3%; the profile does not match

Steelman (own $\pm 5\%$ wage neighbourhood): 19.4% — and the hours block is exactly as inert.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not
become unequal well-being

Responsibility

preference-based choices should not
be compensated

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet, SCW 2005]

$W^1(z, R, A; \mathbf{y}^w)$ — equal pay on the feasible set;
the measure this paper takes to the data

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).